

# AIR QUALITY INDEX

## ANALYSIS



# OUR TEAM



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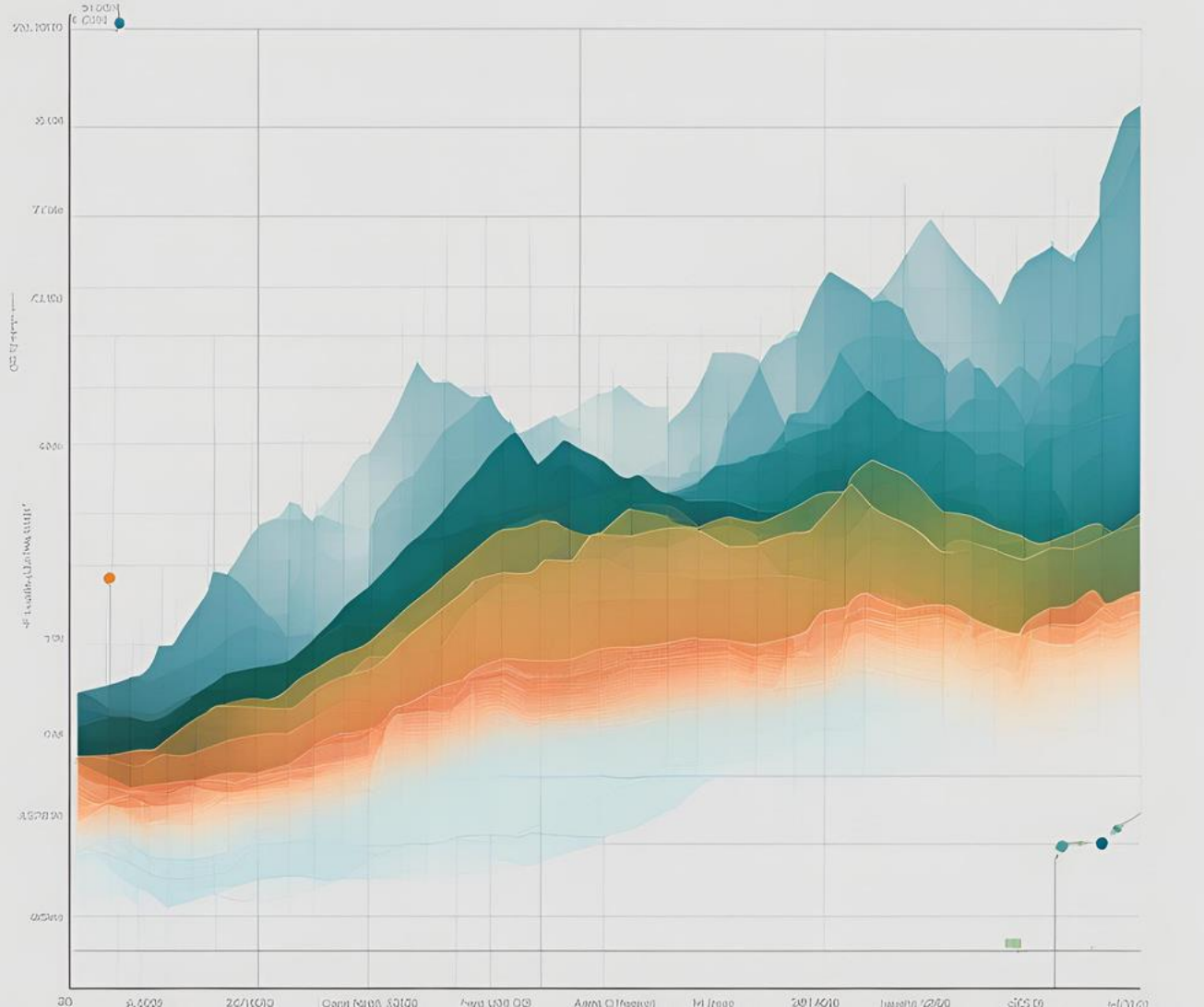
# ABOUT AQI

Air pollution is a critical environmental and health crisis, caused by toxic emissions from vehicles, industries, and human activities. It harms not only the planet but also endangers millions of lives every day. The Air Quality Index (AQI) is a vital tool that highlights the severity of air pollution. Ranging from "Good" to "Hazardous," it warns us of the invisible dangers lurking in the air we breathe.



The project focuses on studying air quality in Indian cities by analyzing PM2.5 and PM10 levels.



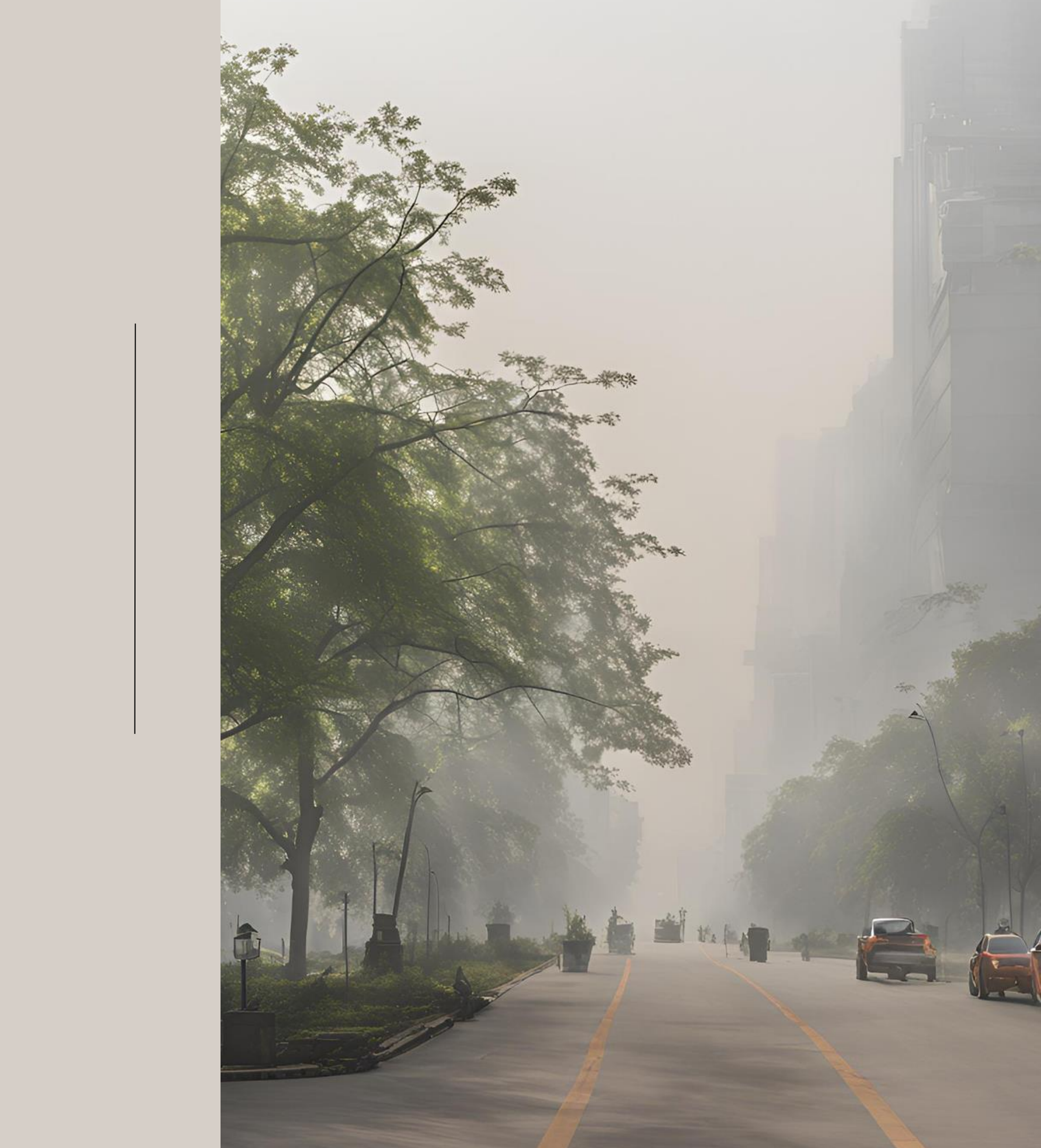


- Allow users to explore and visualize pollution trends for individual cities.
- Provide a user-friendly interface to view and analyze air pollution trends dynamically.
- Offer insights to policymakers and researchers for better urban planning and pollution control measures.

# OUR APPROACH







# DATA COLLECTION

## Data Source:

- It includes critical columns like PM2.5, PM10, AQI and dates.
- Such data might typically come from government agencies, environmental monitoring networks, or publicly available repositories
- Data collection involves loading and cleaning air quality data from a CSV file for analysis.

## Loading Data:

- Reads the CSV file and organizes it into a structured DataFrame.

## Cleaning Data:

- Removes invalid rows and standardizes the date format.

## Aggregating Data:

- Groups by city and month, calculating monthly pollutant averages.

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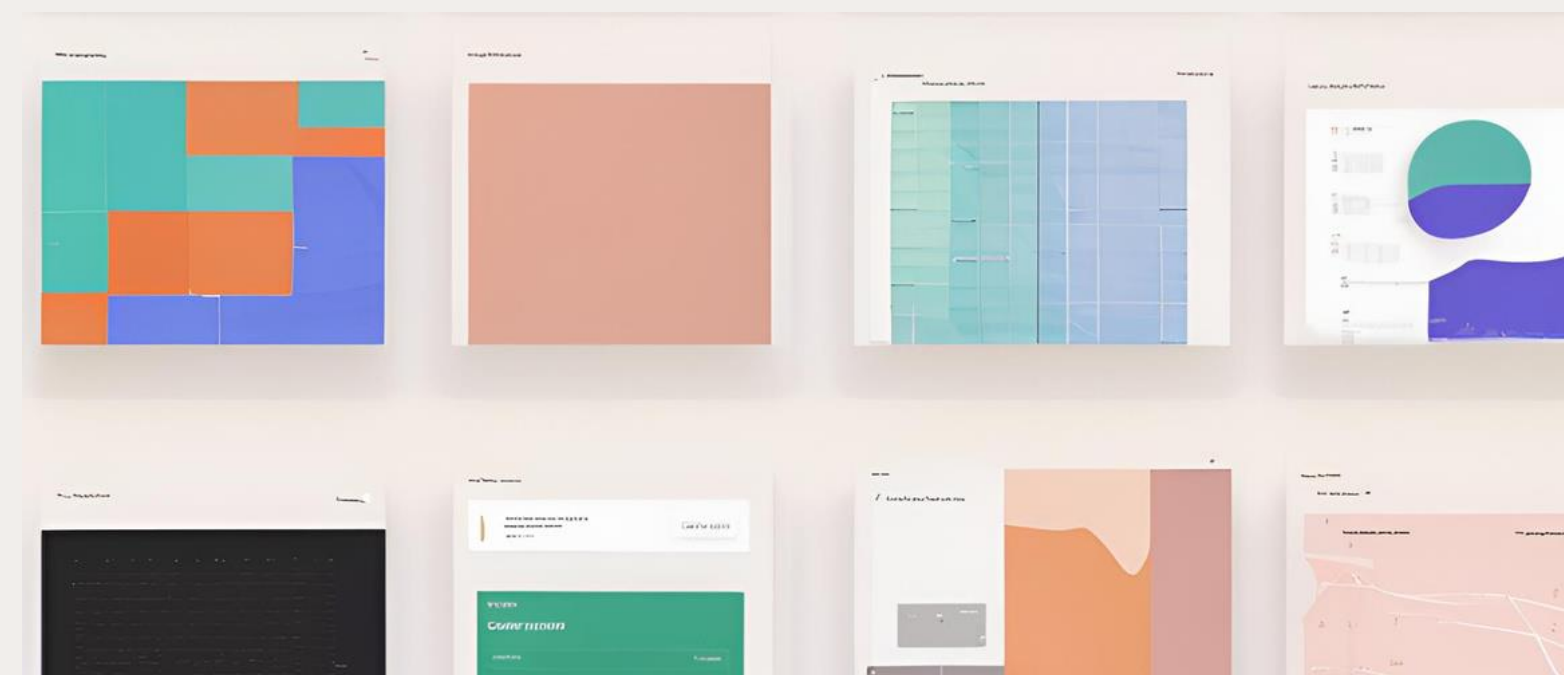
# LIBRARIES USED

## pandas

- Used for data manipulation and analysis.
- Handles loading the CSV file, cleaning the data by removing missing values, and transforming columns (like converting dates to datetime).
- Supports grouping and aggregating data by city and month for trend analysis.

## matplotlib

- A powerful plotting library used to create graphs and visualizations.
- In this code, it helps generate line plots to show PM2.5, PM10, and AQI trends over time.
- Used alongside the FigureCanvasTkAgg backend to integrate plots into the Tkinter GUI.





# LIBRARIES USED

## seaborn

- A data visualization library built on top of Matplotlib.
- Provides aesthetically pleasing, high-level functions for creating statistical plots.
- In this project, it is used for line plots with markers and grid styling to make visual trends clearer.



## tkinter

- Python's built-in library for creating graphical user interfaces.
- Handles the interactive components, such as buttons for loading files, dropdowns for selecting cities, and windows for displaying visualizations.

## matplotlib.backends

- Integrates Matplotlib graphs into the Tkinter GUI.
- Allows plots to be displayed directly in the application window rather than as standalone figures.

(TkAgg)

# CONCLUSION

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## Unveiling the Air We Breathe

Through comprehensive analysis, this project illuminated the state of air quality across multiple cities, uncovering meaningful patterns in PM2.5, PM10, and AQI levels.

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### Key Findings

- Monthly trends revealed critical periods of heightened pollution, offering a roadmap for targeted mitigation efforts.
  - Seasonal variations and consistent patterns provided deeper insights into environmental and urban influences on air quality.
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### A Path Forward

- The AQI bucket trends offer a clear lens into the severity of air pollution, guiding public health strategies.
- Future work can integrate predictive models and larger datasets, paving the way for proactive urban planning and pollution control.





Together, Let's Breathe Better

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# A CALL TO ACTION



This analysis is a step toward fostering awareness, enabling informed decision-making, and promoting sustainable living for cleaner, healthier air.

